The Brent-Price Causal Effect of the 2026 Hormuz Disruption

A Synthetic Control Approach

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BSE, Data Science for Decision Making 2025–2026

From the previous presentation

Original framing. *How does the topology of the global maritime oil trade network shape the dynamics of oil prices?*

- **Q1.** Can network topology improve oil-price forecasts beyond classical time-series baselines?
- **Q2.** How does a maritime disruption propagate into oil prices across horizons, and how long does its effect linger?

Why the pivot to causal SCM.

- Feedback: focus on **causal** estimation rather than forecasting — forecasting is hard and crowded; a credible counterfactual on a specific disruption is more tractable and policy-relevant.
- Reviewed candidate causal methods: event studies, structural VARs (Kilian-style supply/demand decomposition), DiD, network-shock propagation, synthetic control.
- **SCM** selected: cleanest combination of **inferential machinery** (in-space placebo, cross-event weight transfer) and **interpretable result** (donor-weighted counterfactual on the observed price), and it directly produces an ATT estimate rather than a shock decomposition.

Problem statement

Part 1 (today): What was the Brent price impact of the 2026-02-01 Strait of Hormuz disruption?

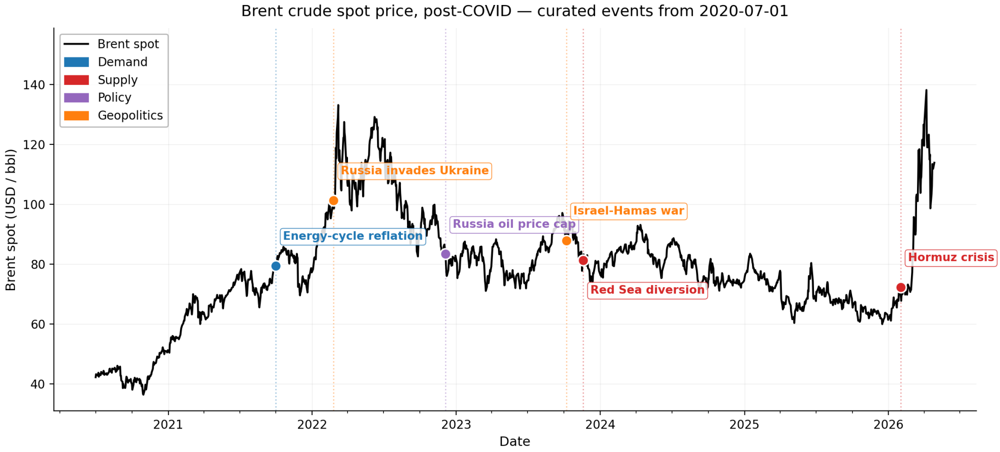
Estimand: ATT over a defined post-event window. Method: synthetic control on the observed price series.

Part 2 (open; discussed at the end).

What is new in the field.

- Closest mechanical SCM-on-price precedent: Austrian retail fuel (Becker, Pfeifer & Schweikert 2021, *Energy Economics*). Field default for oil-price impact: structural VAR (Kilian 2009; Baumeister & Hamilton 2019) or scenario models (Dallas Fed, OIES, Kiel, IEA).
- **Method extension:** retail fuel → crude benchmark; cross-country → cross-asset donors; regulatory → exogenous geopolitical treatment.
- **Two-event robustness design:** Russia 2022 used as a magnitude-validation case — SCM counterfactual within \$3.98/bbl of EIA STEO's last pre-invasion structural forecast on the matched 151-day window ($\sim$ 5.7 pp on implied ATT), an independent-method check that the design generalises to the Hormuz application.

Brent spot price, post-COVID — curated event timeline



Brent spot, post-COVID. Events: demand (blue), supply (red), policy (purple), geopolitics (orange).

Methodology — ensemble of 5 models

Why an ensemble. Headline = median across models; IQR = model-uncertainty band.

Model	What it does	Reference
Convex SCM	Convex weights, no extrapolation	Abadie, Diamond & Hainmueller (2010, *JASA*)
Augmented SCM	Convex + ridge bias correction	Ben-Michael, Feller & Rothstein (2021, *JASA*)
Elastic-net	$L_1 + L_2$ regression, sparse donor weights	Doudchenko & Imbens (2016, NBER WP)
XGBoost	Gradient-boosted trees, flexible nonlinear	Chen & Guestrin (2016, *KDD*)
Bayesian Ridge	Bayesian linear with normal prior, donor PIPs	Brodersen et al. (2015)

Range covered. Convex (no extrap) → augmented (ridge fix) → regression (sparse) → nonlinear → Bayesian (uncertainty).

Model training and validation

	Pre-event (CV)		Treatment (post)	
Russia 2022	2020-07-01 → 2022-02-23	421 obs	2022-02-24 → 2022-09-30	151 obs
Hormuz 2026	2024-06-03 → 2026-01-30	423 obs	2026-02-01 → 2026-05-01	∼59 obs

Training. 5-fold expanding-window walk-forward CV (Hyndman & Athanasopoulos 2018; Bergmeir & Benítez 2012); hparams chosen by pooled val RMSE; final fit refits on full pre-event window.

Validation battery.

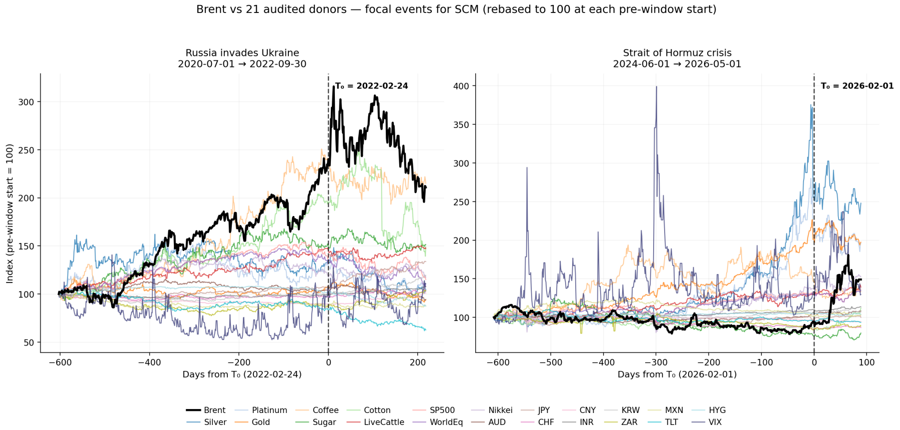
- **Model fit** — walk-forward CV val/train RMSE.
- **SUTVA** — qualitative donor audit on causal channels (Abadie 2021 §3.1).
- **Inference** — in-space placebo, 21- and 33-donor pools (Abadie 2010 §3.3); cross-event weight transfer substitutes for the low-power Hormuz placebo (∼60 post-event obs).

Donor selection

21 audited donors (of 33 candidates). The full 33-donor catalog was audited for SUTVA and pre-event co-movement; 12 were dropped for event-specific exposure (Russia/Ukraine or Persian Gulf channels), leaving 21 in the shared pool used for both events — so we know the full scope of what was excluded and why. Inclusion requires *both*: **pre-treatment** co-movement with Brent through a common global factor, *and* **treatment** cleanliness (SUTVA).

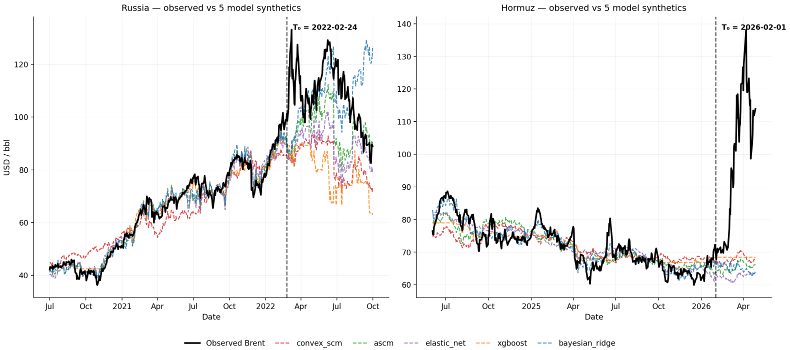
Category	Chosen donors	Factor channel (pre) & cleanliness (treatment)
Metals (3)	Silver, Platinum, Gold	Real rates / safe-haven / auto industry; no Russia or Gulf supply routing.
Agriculturals (4)	Coffee, Sugar, Cotton, LiveCattle	Global trade activity (Brazil / India / US producers); outside the Russia-Ukraine grain belt.
Equities (3)	SP500, WorldEq, Nikkei	Global growth / equity risk premium; no direct Russia or Gulf-energy exposure.
FX (8)	AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN	Safe-haven, EM growth, China demand; no petrocurrency or EU-energy-crisis loading.
Rates / credit (2)	TLT, HYG	US duration / credit-risk premia; not directly oil-linked.
Volatility (1)	VIX	S&P 500 implied vol — risk-on/off via equities.

Donor co-movement with Brent — audited pool



Brent (black) vs 21 audited donors, rebased to 100 at pre-window start; dashed line marks T_0. Same windows as the SCM.

Validation results — model fit



	Russia 2022			Hormuz 2026		
Model	**train**	**val**	**val/train**	**train**	**val**	**val/train**
Convex SCM	0.109	0.127	1.16	0.059	0.075	1.26
ASCM	0.043	0.088	**2.04**	0.047	0.064	1.35
Elastic-net	0.049	0.089	1.83	0.045	0.070	1.55
XGBoost	0.039	0.117	**3.00**	0.044	0.076	1.72
Bayesian Ridge	0.031	0.096	**3.07**	0.031	0.066	**2.10**

Validation results — in-space placebo

Abadie 2010 §3.3: rank Brent's post/pre RMSPE ratio against placebo refits. 21-donor floor = 0.045; 33-donor full pool re-run gives floor = 0.029.

	Russia 2022		Hormuz 2026	
Model	21-donor	33-donor	21-donor	33-donor
Convex SCM	0.545	0.500	**0.045**	**0.029**
ASCM	0.682	0.471	**0.045**	**0.029**
Elastic-net	0.409	0.412	**0.045**	**0.029**
XGBoost	**0.045**	0.147	**0.045**	**0.029**
Bayesian Ridge	0.636	0.529	**0.045**	**0.029**

Headline.

- **Hormuz:** all 5 models hit the floor under both pools.
- **Russia:** not decisively rejected — only XGBoost reaches floor at 21-donor and fails at 33-donor. Feb-22 macro regime shift moves many donors too; the historical \$95–\$130 record, EIA STEO cross-check, and cross-event transfer (§5f) carry the Russia case.

External validation — EIA STEO pre-invasion forecasts (Russia only)

Cross-method check. SCM counterfactual vs EIA's last pre-invasion Brent forecast (STEO Feb-22, issued 2022-02-08). No shared information path (SCM uses cross-assetco-movement; STEO uses fundamental supply-demand modelling)

Model	**Synth \$/bbl**	**STEO Feb-22 \$/bbl**	**Δ**
Convex SCM	83.55	84.94	−\$1.39
ASCM	95.52	84.94	+\$10.58
Elastic-net	88.92	84.94	+\$3.98
XGBoost	79.28	84.94	−\$5.66
Bayesian Ridge	107.91	84.94	+\$22.97 (deg.)
Ensemble median	**88.92**	**84.94**	**+\$3.98**

Implied ATT (Brent actual \$108.54): SCM ensemble +22.1% vs STEO Feb-22 +27.8% (Δ ∼ 5.7 pp).

Headline result — ATT estimate

Why ATT? One treated unit (Brent) → effect *for that unit*, not a population ATE (Abadie 2010, 2021). Mean gap over the post-event window — integrates path-reversion, robust to single-day liquidity jumps.

Model	**Russia 2022**	**Hormuz 2026**
Convex SCM	+29.6%	+38.6%
ASCM	+13.3%	+43.7%
Elastic-net	+21.7%	+49.8%
XGBoost	+36.8%	+36.8%
Bayesian Ridge	+0.5% (deg.)	+43.6%
Ensemble median	+21.7%	+43.6%
IQR	[+13.3, +29.6]	[+38.6, +43.7]

Reading. Russia +21.7% within ∼5.7 pp of STEO Feb-22 ATT (synth \$88.92 vs STEO \$84.94, Δ \$3.98/bbl); Bayesian Ridge degenerate, IQR excludes it. Hormuz +43.6% with tight IQR — place against Dallas Fed / OIES / IEA scenarios.

Part 2 — ideas for the second half

[to be added by author]

Possible directions: pass-through to retail fuel / headline CPI; SCM vs scenario-model comparison; severity-dependent extrapolation; alternative donor pools.

Thank you

Questions and discussion

github.com/ElvisCasco/Brent_Analysis

Appendix — excluded donors

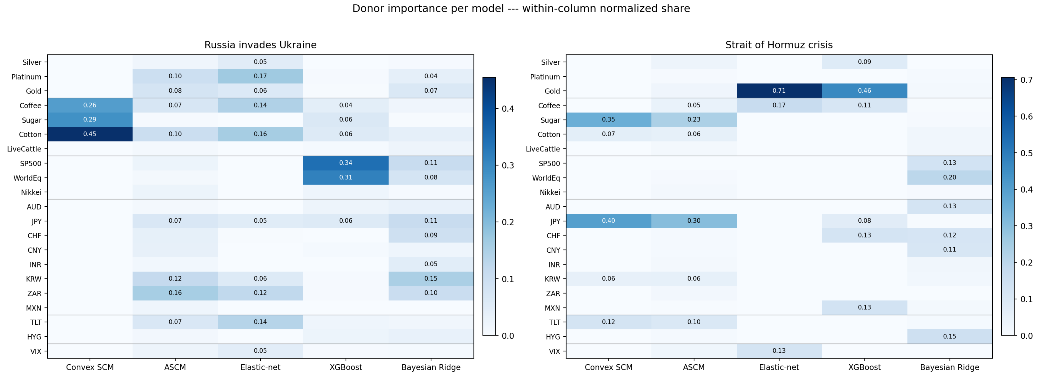
A. Categorical exclusions (never enter the candidate pool).

Category	Excluded & reason
Energy complex	WTI, NatGas, refined products (RBOB, heating oil, jet fuel), EIA inventories — mechanical SUTVA violation on any oil shock.
Petrocurrencies	CAD, NOK, BRL, RUB, COP, MYR — value partly set by oil exports.
FX unstable	TRY, ARS, EGP — domestic crisis dynamics dominate; noise, not signal.
Equities rejected	DAX, FTSE, TSX 60 (energy weight), KOSPI, Hang Seng (redundant with KRW / CNY).
Commodities rejected	Cocoa, OJ, Lean hogs (thin / idiosyncratic), Lumber (LBS→LBR contract break), Aluminium (Rusal ∼6% supply).

B. Per-event SUTVA exclusions (33-donor pool → 21-donor strict-clean preferred).

- **Russia H-tier** (direct supply/demand): Wheat, Corn, Palladium, EUR, DXY, BTC.
- **Russia M-tier** (terms-of-trade / secondary): Copper, IronOre, Soybeans, EM_Eq, GBP, US10Y.
- **Hormuz:** no H/M flags — audit clean on all 33 candidates. The 21-donor intersection is used to enable cross-event weight transfer (§5f), at the cost of 12 Hormuz-clean donors.

Appendix — donor importance



Per-model raw metric: convex weight w_j (SCM, ASCM) / $|\hat{\beta}_j|$ (Elastic-net, Bayesian Ridge) / feature importance (XGBoost). Cells = $|w_j|$ normalised within each column to sum to 1.

Appendix — donor SUTVA cleanliness

Permutation mean-shift test at known T_0 on donor log-returns (Chow 1960 hypothesis + Lehmann & Romano 2005 permutation + Phipson & Smyth 2010 correction). Combined flag = permutation *or* Wilcoxon event-window rejects under Benjamini-Hochberg FDR ($\alpha = 0.10$).

	Permut.	**Event-wind.**	**Combined**	**Audit**
Russia 2022	1 / 33	0 / 33	1 / 33	12 / 33
Hormuz 2026	0 / 33	1 / 33	1 / 33	0 / 33

- **Hormuz: 32 / 33 agree.** Audit's "no Persian-Gulf routing" confirmed; only IronOre triggers (event-window).
- **Russia: 20 / 33 agree.** Audit's H/M tier captures *causal channels* (Wheat: Russia+Ukraine $\approx$ 30% world exports) that daily-return tests can't reproduce at $N = 33$ after FDR. Tests' lone rejection (CNY) is China zero-COVID, not Russia-treatment.

Appendix — in-time placebo & LOO

In-time placebo (Abadie 2015) — mean fake-period gap (%) at $T_0^{\text{fake}} = \text{real } T_0 - 6$ months, defaults hparams. **LOO** — range (max − min, pp) across leave-one-donor-out refits.

	In-time gap (%)		LOO range (pp)	
Model	**Russia**	**Hormuz**	**Russia**	**Hormuz**
Convex SCM	+1.9	−11.9	6.6	9.4
ASCM	−4.2	−9.8	13.4	8.9
Elastic-net	+7.5	−10.6	15.6	11.0
XGBoost	+26.4	−12.4	**4.9**	**4.7**
Bayesian Ridge	−5.3	−12.2	24.9 †	13.1

In-time. Russia fake window contains late-2021 oil rally + Jan-2022 invasion runup — partial real-effect contamination expected; XGBoost +26% low-power (training shrinks $\sim$ 31%). Hormuz −10 to −12% across all five models = pre-Hormuz runup donors did not absorb, not model failure.

LOO. Tightest: **XGBoost** ($\sim$ 5 pp, bold). Widest: **Bayesian Ridge Russia** († range crosses zero, −12.8 to +12.0) — LOO-fragile, consistent with its 3.07 walk-forward overfit ratio and 0.5% baseline. Ensemble median robust because Convex / ASCM / EN / XGBoost agree more closely.

Appendix — cross-event weight transfer

§5f — strongest single test of methodology generalization. Russia-fitted weights applied to the Hormuz post-window vs independently-fit Hormuz gap:

Model	Independent	Transferred	Δ
Convex SCM	+38.6%	+47.0%	+8.4 pp
ASCM	+43.7%	+27.3%	−16.4 pp
Elastic-net	+49.8%	+23.2%	−26.6 pp
XGBoost	+36.8%	−58.2%	−95.0 pp (deg.)
Bayesian Ridge	+43.6%	−100.0%	−143.6 pp (deg.)

Reading.

- Convex SCM transfers cleanly (+8 pp drift).
- ASCM and Elastic-net give qualitatively consistent direction with magnitude attenuation.
- XGBoost and Bayesian Ridge degenerate — regime drift in Russia-fitted weights blows up the projection.
- Factor structure has shifted between 2020–22 and 2024–26, but the **sign and order of magnitude** of the Hormuz effect are robust under the linear-convex models.